



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Vishnupuri, Nanded, Maharashtra, India**

ISE 1 (Semester-I) 2025-26

Department: First year

F.Y. B. Tech

Programming for Problem

Solving ESC--103

Date:16/10/2025 Time:11:00am

Max. Marks :20

- Note: i. Attempt any 4 questions
ii. Assume suitable data if necessary

Q.N.	Question	Marks	CO	BT
Q.1	What is a variable in C programming? Explain the different data types used to declare variables with example.	5	CO1	BT1
Q.2	Write a C program to find the largest number among three numbers.	5	CO4 CO5	BT4
Q.3	Differentiate between while, do-while, and for loops using C Program.	5	CO3	BT2
Q.4	What is a flowchart? Draw a flowchart to find positive and negative number provided by user as input.	5	CO2	BT3
Q.5	Explain assignment and logical operators with programming example?	5	CO1	BT2

Course Outcomes (CO)

CO1. To formulate simple algorithms for arithmetic and logical problems.

CO2. To translate the algorithms to programs (in C language).

CO3. To test and execute the programs and correct syntax and logical errors.

CO4. To implement conditional branching, iteration, and recursion.

CO5. To decompose a problem into functions and synthesize a complete program

using divide and conquer approach.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

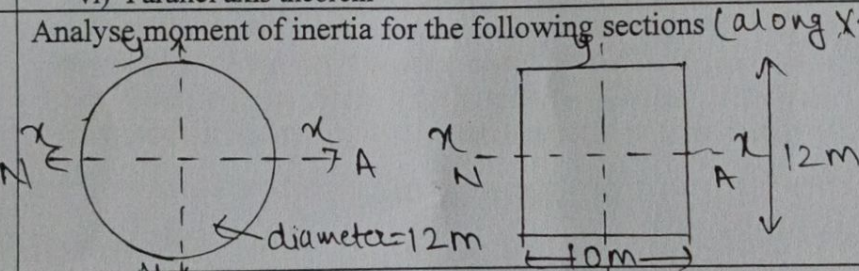
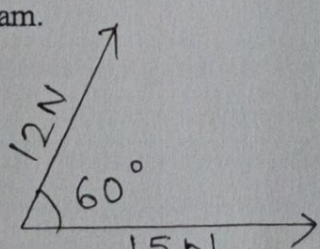
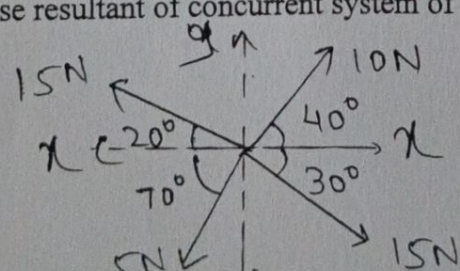


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of BSHM
In Semester Examination I
Academic Year: 2025-26

Class: FY	Subject: Engineering Mechanics	Subject Code: ESC 109
Date: 16.10.2025	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. All questions are compulsory

Q.N.		Marks	CO	BT
Q.1	a Explain the following i) Lami's Theorem ii) Varignon's theorem iii) Centroid vi) Parallel axis theorem	8	CO4	BT1
Q.2	a Analyse moment of inertia for the following sections (along X-X) axis 	4	CO2	BT4
Q.3	a Analyse resultant of system of forces which denotes 2 sides of parallelogram. 	4	CO2	BT4
Q.4	a Analyse resultant of concurrent system of forces 	4	Co 1	BT4

== ALL THE BEST ==

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	Academic Year: 2025-26	Semester: I	Examination: ISE-I
	Class: First Year	Branch: Basic Science	
	Date: 17/10/2025	Time: 11:00 - 11:45	
Subject: Biology for Engineers		Code: BSC105	Maximum Marks: 20

General Instructions:

- I. This paper contains 1 (One) page. Figures to right indicate full marks.
- II. Answer any four question.
- III. Draw neat & compact sketches wherever necessary.
- IV. Assume data wherever required and state them clearly.

Que.	Question	Marks	CO	BT
1.	Explain Enzymes and their application in the industries.	5	CO2	BT2
2.	Examine an animal and plant cell. List out the differences.	5	CO1	BT4
3.	Write a detailed note on RNA and classify them.	5	CO2	BT4
4.	Write the characterisitscs of prokaryotic cell with neat sketch.	5	CO1	BT1
5.	Explain prophase-I of meiosis in detail.	5	CO1	BT2

**** ALL THE BEST ****

CO1: Introduction to basics of biology which includes cell, the unit of life, different types of cells, classification of living organisms.

CO2: Understanding what are biomolecules present in a cell, the structures function, and their role in a living organism.

CO3: Application of a certain biomolecules in an industry.

CO4: How biology can be applied in our daily life using different technology for production of medicines to transgenic plants and animals to designing new biological products.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.



Shri Guru Gobind Singhji Institute of Engineering & Technology,
Nanded
Department of Electronics and Telecommunication Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: FY B. Tech. Subject: Basic Electronics

Couse Code: (PCC-EC/EL/IN 102; PCC-EC 103; ESC 104)

Date: 14/10/2025 Time: 11.00-11.45

Max. Marks: 20

Q.N.	All questions are compulsory.	Mark	CO	BT
Q.1	A P-N junction diode having internal resistance $r_f = 20 \Omega$ is used for half wave rectification. If the applied voltage $v = 50 \sin \omega t$ and load resistance $R_L = 800 \Omega$, find 1. I_m 2. I_{dc} 3. a.c. power input and d.c. power output 4. d.c. output voltage 5. Efficiency of rectification.	10	CO4	BT5
Q.2	Draw and explain bridge full wave rectifier with input and output waveforms in detail.	05	CO3	BT3
Q.3	Define the following a. Peak Inverse Voltage (PIV) b. Ripple factor c. Reverse current or leakage current d. Diode Equation e. Efficiency of rectifier	05	CO1	BT2

Bloom's Taxonomy (BT)

BT1-Remember, BT2 -Understand, BT3 -Apply, BT4 -Analyze, BT5 -Evaluate, BT6 -Create

Course Outcomes:

CO1	To introduce some of the basic electronic components and circuits.
CO2	To study the characteristics and working of BJT.
CO3	To explain the applications of components and devices.
CO4	To understand the concept of transistor biasing and its applications.
CO5	To know oscillators and operational amplifiers.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of BSH&M

Academic Year: 2025-26 (semester I)

In-Semester Examination / Evaluation I

Class: First Year B. Tech.

Course Name: Engineering
Thermodynamics and Mechanics

Course Code: PCC-ME 103
ESC 105

Date: 16.10.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones are strictly prohibited.
- iii. Non-programmable calculators are allowed.
- iv. Exchange/Sharing of stationery, calculator etc. not allowed.
- v. Figures to right indicate full marks.
- vi. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	State and explain the statements of first law of thermodynamics with its limitations and PMM-I.	6	CO1	BT2
Q.2	In a gas turbine, the gases flow at the rate of 5kg/s. The gases enter the turbine at a pressure 8 bar with a velocity 120 m/s and leaves at a pressure 3 bar with velocity 250 m/s. The turbine is insulated. If the enthalpy of the gas at inlet is 800 kJ/kg and at outlet 500 kJ/kg, determine the capacity of the turbine.	8	CO1	BT4
Q.3	Explain the similarities and dissimilarities between heat and work.	6	CO1	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

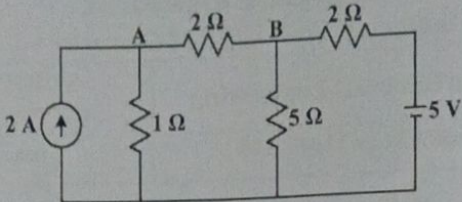
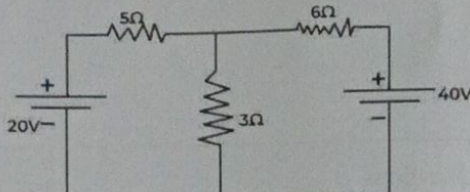
	On successful completion of this course, students should be able to:
CO 1	Apply the thermodynamic laws to analyse the thermodynamic systems.
CO 2	Illustrate various basic parts and transmission systems.
CO 3	Understand the need and steps of machine design.
CO 4	Determine resultant of various force systems.
CO 5	Apply the principles of static and dynamic mechanics to solve problems related to forces, equilibrium, motion, and friction.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering In-Semester Examination I Academic Year: 2025-26 (Semester – I)	
Class: F.Y (GENERAL)	Subject: Basic Electrical Engineering	SUBJECT CODE- ESC101/105/PCC EL103
Date: 15.10.2025	Time: 11:00AM to 11:45 AM	Max. Marks: 20

Note:

- i. *All Questions are compulsory.*
- ii. *Figures to right indicate full marks.*
- iii. *Question paper contains only 02 pages.(See backside of paper)*

Q.N.	Questions	Marks	CO1	BT
Q.1	a) Write short note on types of sources with symbols . (2 mark) b) State Kirchhoff's both law's. (2 mark)	4	CO-1	BT-1
Q.2	a) For the circuit shown in below Calculate the resistance AB using star delta transformation (4 mark) b) Two resistors $R_1 = 2\ \Omega$ and $R_2 = 8\ \Omega$ are connected in parallel. The total current entering the parallel combination is 10 A. Calculate the current through R_1 using the current division rule. (2 mark)	6	CO-2	BT3

Q.3	Calculate the current through the branch AB for the circuit shown in circuit using Nodal Analysis 	5	CO-2	BT-3
Q.4	Calculate the Current in 3 ohm Resistor by using superposition theorem. 	5	CO-2	BT-3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Describe the basic concepts of voltage, current, EMF, energy sources, and fundamental laws like Ohm's and Kirchhoff's in electric circuits. (BT01 – Understand)
CO 2	Analyze DC electrical circuits using series-parallel combinations, mesh and nodal analysis, and theorems such as Superposition, Thevenin's, and Norton's. (BT04 – Analyze)
CO 3	Apply electromagnetic laws to analyse magnetic circuits, induced EMF, and energy storage in magnetic fields. (BT03 – Apply)
CO4	Analyze single-phase and three-phase AC circuits using phasor diagrams, and calculate impedance, power factor, and power components. (BT04 – Analyze)
CO5	Explain the construction and operation of single-phase transformers and basic electrical measuring instruments like PMMC meters and wattmeters. (BT01 – Understand)



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

Department: Department of Basic Sciences, Humanities & Management

In Semester Examination-I

Academic Year: 2025-26 (Semester-I)

Class: F.Y. B. Tech.	Subject: Chemistry	Subject Code: BSC102
Date: 13/10/2025	Time: 4.30 pm - 5.15 pm	Max. Marks: 20

Note:

- Assume suitable data if necessary and state clearly
- Use of scientific calculator is allowed.
- Exchange/sharing of stationery, calculator etc. not allowed
- The symbols carry the usual meaning.

Data: Atomic Weight: Ca = 40, H = 1, O = 16, Cl = 35.5, S = 32, Mg = 24, N = 14, Si = 28, C = 12, Na = 23

Q. 1	Answer any three from the following	Marks	CO	BT
a	Calculate carbonate & non carbonate hardness of water containing following impurities in mg/Lit. $MgH(CO_3)_2 = 7.1$, $CaH(CO_3)_2 = 8.1$, $MgCO_3 = 4.2$, $CaCO_3 = 10$, $MgSO_4 = 24$	3	CO1	BT3
b	Explain the principle of EDTA method.	3	CO4	BT2
c	Define Monomer, Polymer, Polymerization give example of each.	3	CO3	BT1
d	Polymers are being used in the field of medicine & surgery. Which unique properties of polymers made this possible?	3	CO2	BT2
Q. 2	Solve any one question			
a	The hardness of 100000 liters of hard water sample was completely removed by passing it through a zeolite softener. The zeolite softener requires 400 liters of sodium chloride solution containing 100 g of NaCl per liter for regeneration. Determine hardness of water sample.	5	CO5	BT3
	OR			
b	Explain preparation method, Properties & Applications of PMMA	5	CO1	BT1
Q. 3	Solve any one question			
a	1.2 gm of $CaCO_3$ was dissolved in HCl and the solution was made up to 1 liter with distilled water. 20 ml of the solution required 30 ml of EDTA solution for titration. 50 ml of hard water sample required 25 ml of EDTA. 100 ml of hard water sample after boiling and filtering required 25 ml of EDTA solution. Calculate each type of hardness of water.	6	CO4	BT4
	OR			
b	What is vulcanization? explain it along with the reaction and give the advantages of vulcanized rubber over natural rubber (any 8 points).	6	CO3	BT2

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

Course Outcomes

	On successful completion of course, students should be able to:
CO1	Students will be able to understand various industrial processes and products.
CO2	Enabling students to know about the relevant engineering materials, their chemistry and recent technology.
CO3	Outer world will be benefited with current scientific knowledge.
CO4	Students will be able to understand various analytical methods used for material analysis.
CO5	To strengthen the student's understanding to the related core subjects in their respective discipline.

	Shri Gurb Gobind Singhji Institute of Engineering and Technology, Nanded Department of BSH&M In-Semester Examination I Academic Year: 2025-26 (Semester – I)	
Class: FY B.Tech.	Subject: Professional Communication	Subject Code: AES102/HMC110
Date: 15.10.2025	Time: 04:30 to 05:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry usual meaning.

Q. No.	Attempt all the following questions as per the instruction given.	Marks	CO	BT
Q.I	Do as directed. (Any ten)	10	CO1	BT1
1.	This is the worst example of dishonesty that I have ever come across. (Identify the sentence structure)			
2.	Deepak took leave of me. (Identify the sentence structure)			
3.	We often find things <u>of this kind</u> . (Identify the underlined parts of speech)			
4.	Many persons, ---- we know are hypocrites. (Use appropriate pronoun)			
5.	Your mother, and not your father <u>was present</u> . (Identify the underlined parts of speech)			
6.	The clothes did not dry <u>as it was raining</u> all night. (Identify the underlined parts of speech)			
7.	He was determined to fight <u>to the bitterest end for the sake of justice</u> . (Identify the underlined parts of speech)			
8.	Oh! I didn't know you were coming to preside. (Identify the tense)			
9.	By next year he will have been working in this factory for five years. (Identify the tense)			
10.	Dinesh --- no pen to write. (Use appropriate verb form)			
11.	I --- how it was made. (wonder)			

Q.II	Change the voice of the following sentences.	05	CO2	BT2 & BT3
1.	Who stole your bag from the bus?			
2.	Close the door.			
3.	The teacher has not yet evaluated the answer books.			
4.	The police will have arrested the thief.			
5.	The students were given intense training in spoken English.			
6.	May I help you?			
Q.III	Transcribe the following into IPA symbols. (Any five)	05	CO2	BT2
1.	three			
2.	house			
3.	cheese			
4.	see			
5.	love			
6.	dad			
7.	where			

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 – Remember, BT2 – Understand, BT3 – Apply, BT4 – Analyze, BT5 – Evaluate, BT6 – Create
Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To enable students to learn the structure and style of effective sentences.
CO 2	To enable students to express themselves in English effectively.
CO 3	To enhance students' Listening, Speaking, Reading & Writing Skills.
CO 4	To help students to understand nuances of technical writing.
CO 5	To communicate in proper accepted layout formats with efficiency.